

CORRUPTION & SECURITY ANALYSIS PROJECT

Statistical Modelling of Perceived Corruption and Security

Description:

This project applies statistical modelling techniques to examine the relationship between perceived corruption and security using survey data from Hong Kong.

The analysis focuses on identifying associations between corruption perception, demographic characteristics, and precautionary behaviour, as well as testing potential interaction effects (moderation) between variables such as confidence in the police.

Methods include binary logistic regression, multiple linear regression, and moderation analysis to provide a structured, data-driven understanding of how subjective corruption influences real-world behaviour.

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Field: Data Analytics / Financial Crime / Criminology

Tools: R (Statistical Modelling), Regression Analysis

Dataset: World Values Survey (Hong Kong, ~2000 observations)

Year: 2024

EXECUTIVE SUMMARY

Objective: Analyse the relationship between perceived corruption and security behaviour using survey data.

Data: World Values Survey (Hong Kong, ~2000 observations)

Methods:

- Binary logistic regression
- Multiple linear regression
- Moderation analysis

Key findings:

- Higher perceived corruption → increased likelihood of security behaviour (+10% odds)
- Demographic variables influence corruption perception.
- No significant interaction effect with confidence in police.

BACKGROUND

This analysis uses data from World Value Survey (WVS). The goal of the WVS is to study and provide an understanding of attitudes and believes of people across different societies. For the Hong Kong dataset a mixed-methods methodology was used: face-to-face survey and online survey. Overall, 2075 participants took part. Data was collected approximately over the period of 4 months (from July 2018 to November 2018). The strengths and weaknesses of the dataset were derived from WVS7 Methodology Report Hong Kong 2018. The first strength is that participants came from diverse demographic backgrounds. For example, after the data was weighted, it showed that 52.7% of participants were female and 47.3% were male, which demonstrates that there was no underrepresentation. Furthermore, the same goes for other categories such as age groups and education levels. However, one limitation is that participants were not assessed based on their income, which could potentially mean that there was underrepresentation of certain groups of people (such as groups of very low or very high income). The second strength is that for the face-to-face survey, there was no upper age cut-off, which allowed opinions of older population to be heard. However, an upper age cut-off for the online survey was 65 years, which is a limitation. Furthermore, for both face-to-face survey and online survey participants had to be 18 years or over, which is a limitation as voices of younger members of the community were not heard.

METHODOLOGY

Variables chosen for analysis:

	Variable used	Answers	Variable type	Why chosen
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Corruption	Perceptions of corruption in the country	scale of 1-10 (1 – no corruption, 10 – abundance of corruption)	Ordinal	Assess whether corruption is a problem in Hong Kong and what action needs to be taken
Security	Things done for reasons of security: Didn't carry much money	Yes/ no	Binary (nominal)	Assess how safe citizens are feeling and apply new policies to provide security
Sex	Respondent's sex	Male/ female	Binary (nominal)	Targeted policy applications to certain demographic groups
Age	Age of respondent	XX	Ratio	
Immigration status	Respondent immigrant	I am born in this country/ I am an immigrant to this country (born outside this country)	Binary (nominal)	
Confidence in police	How confident respondent is in the police	scale of 1-10 (1 – most confident, 10 – not confident at all)	Ordinal	To assess to what extent respondents trust police and what can be done to improve trust

This report will answer 4 statistical research questions (RQ):

1. Is there an association between the perception of security and subjective corruption in the country?

Binary logistic regression was chosen as an analytical approach to answer this question because two variables need to be assessed, one binary (security, outcome) and one ordinal (corruption, explanatory).

2. Is there an association between demographic characteristics and people's subjective corruption and perception of security?

This question looks at two different associations; thus, two models were chosen to analyze it. Multiple linear regression was most appropriate to analyse the association between demographic variables and corruption because it has one outcome variable and multiple explanatory variables. Binary logistic regression was chosen to analyze the association between demographic variables and perception of security because it allows to look at associations between one binary outcome variable and multiple explanatory variables.

3. Using all demographic variables as in (2), is the association between the perceived security and subjective corruption dependent on an individual's confidence in the police?

Multiple linear regression with moderation was most appropriate for this question. It allows to assess whether the relationship between one explanatory variable and the outcome variable depends on the value of another explanatory variable.

4. Is there an association between confidence in the police and age?

This question only has two variables to analyse, none of which are binary. Since variables in question are ordinal and ratio, simple linear regression was assumed most appropriate.

RESULTS & INTERPRETATION

RQ1

There is an association between the perception of security and subjective corruption in Hong Kong. All else being equal, a one-unit increase in perceiving corruption as being high increases the odds of not carrying much money due to security reasons by 10%. 95% of the estimated true values lie between 1.06 and 1.15, indicating that the association is significant, as confidence intervals do not include the number 1.

RQ2

After fitting a multiple linear regression following associations between perception of corruption and demographic variables were identified:

- All else being equal, being female is partially associated with a 0.09 decrease in perceiving corruption as high compared to being male. However, this partial association is not significant because p-value is bigger than 0.05.
- Ceteris paribus, a one-unit increase in age is partially associated with an estimated 0.008 decrease in perceiving corruption as high. The partial association is significant because p-value is smaller than 0.05.

- Controlling for all other variables, being an immigrant in Hong Kong is partially associated with a 0.62 decrease in perceiving corruption as high compared to not being an immigrant. This association is significant as p-value is lower than 0.05.

Looking at the adjusted R^2 it is evident that 0.2% of the variance in the outcome variable can be explained by the model. Based on the F-statistic and the p-value null hypothesis that no model is better than the existing model can be rejected because both values are above the threshold for significance.

Fitting a binary logistic regression to assess the correlation between security perception and demographic variables resulted in the following conclusions:

- All else being equal, being female significantly increases the odds of not carrying much money due to security reasons by 94%. 95% of the estimated true values lie between 1.62 and 2.33, indicating that the association is significant, as confidence intervals do not include the number 1.
- Controlling for all other variables, a one-unit increase in age increases the odds of not carrying much money due to security reasons by 0.6%. 95% of the estimated true values lie between 1.00 and 1.01, which indicates that the association is not significant, as confidence intervals include the number 1.
- Ceteris paribus, being an immigrant in Hong Kong decreases the odds of not carrying much money due to security reasons by 3%. 95% of the estimated true values lie between 0.78 and 1.19, which indicates that the association is not significant, as confidence intervals include the number 1.

RQ3

All else being equal, the interaction term **corruption:confidence** in the police is not significant because the p-value is 0.51, which is above the threshold for significance (0.05). It can be concluded that the perceived security and individual's confidence in the police do not influence each other. Consequently, since perception of corruption is an outcome variable in this model it is evident that the association between perceived security and subjective corruption is not dependent on individual's confidence in the police.

RQ4

There is an association between confidence in the police and age. A one-unit increase in age is associated with an estimated 0.011 increase in having less confidence in the police. This is relevant because of the p-value, which is less than 0.05. Approximately 0.06 variance in the outcome variable can be explained by the model.

	Explanatory variables	Model 1 (Outcome: security)	Model 2 (Outcome: corruption)	Model 3 (Outcome: security)	Model 4 (Outcome: corruption)	Model 5 (Outcome: confidence in the police)
Explanatory variables	Perception of corruption	0.099 * [0.054; 0.143]				
	Perception of security				0.609 [-0.044; 1.262]	
	Sex		-0.093 [-0.268; 0.082]	0.663 * [0.481; 0.844]	-0.163 [-0.333; 0.007]	
	Age		-0.008 * [-0.014; -0.003]	0.006 [-0.000; 0.011]	-0.002 [-0.008; 0.003]	0.011 * [0.009; 0.013]
	Immigrant		-0.616 * [-0.820; -0.412]	-0.034 [-0.246; 0.178]	-0.382 * [-0.582; -0.182]	
	Confidence in the police				-0.661 * [-0.851; -0.472]	
	perception of corruption: confidence in the police				-0.079 [-0.310; 0.153]	
Model intercept	constant	-0.003 [-0.257; 0.252]	6.753 * [6.317; 7.190]		7.825 * [7.207; 8.442]	2.175 * [2.076; 2.275]
Model fit and goodness of fit	AIC	2698.884		2664.957		
	BIC	2710.143		2687.473		
	R^2		0.026		0.101	0.055
	Adjusted R^2		0.024		0.098	0.055
	F-statistic		18.06		37.93	119.3
Number of observations	N	2058	2042	2057	2035	2049

Table 1: Regression Results (Model 1-Model 5)

*p<0.05 [95% confidence Intervals]

DISCUSSION

Since people with high perception of subjective corruption are feeling less secured, one approach to enhance citizens' sense of security in Hong Kong is to reduce corruption. However, according to Transparency International Hong Kong is already

one of the least corrupt countries in the world (ranking 12th out of 180). Since the variable that was used to assess corruption in this report was subjective, one way to address it could be holding campaigns dedicated to confronting corruption. People seeing that the government is working on combatting corruption can decrease the overall subjective perception of corruption, consequently making people feel more secure. Furthermore, campaigns can be designed to address certain segments of the population: specifically younger population and non-immigrants, as older population and immigrants tend not to assess corruption as high.

BIBLIOGRAPHY

1. World Value Survey. Wave 7 (2017-2022).
<https://www.worldvaluessurvey.org/WVSDocumentationWV7.jsp>
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